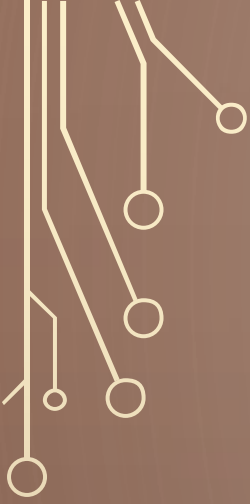




HABOTCONNECT HIRING PROJECT

SECURE CLOUD INFRASTRUCTURE PROVISIONING
USING TERRAFORM, GOOGLE CLOUD PLATFORM,
GITHUB ACTIONS & DJANGO REST FRAMEWORK

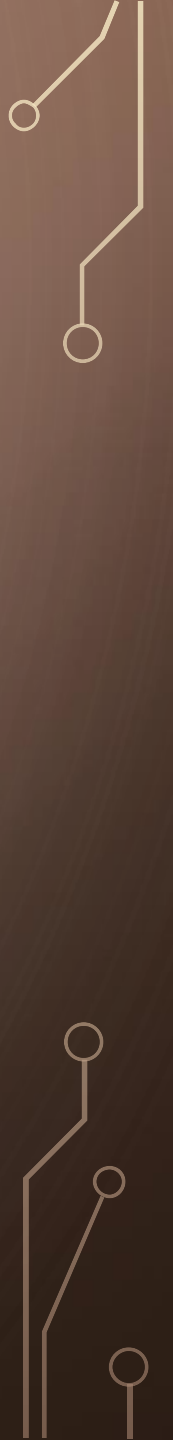
AMRUTA SURYAVANSHI



PROJECT OVERVIEW

- • Provision secure GCP staging infrastructure
- Automate deployment using Terraform
- Protect student data
- Implement Fail-Closed CI/CD

PROJECT OBJECTIVES

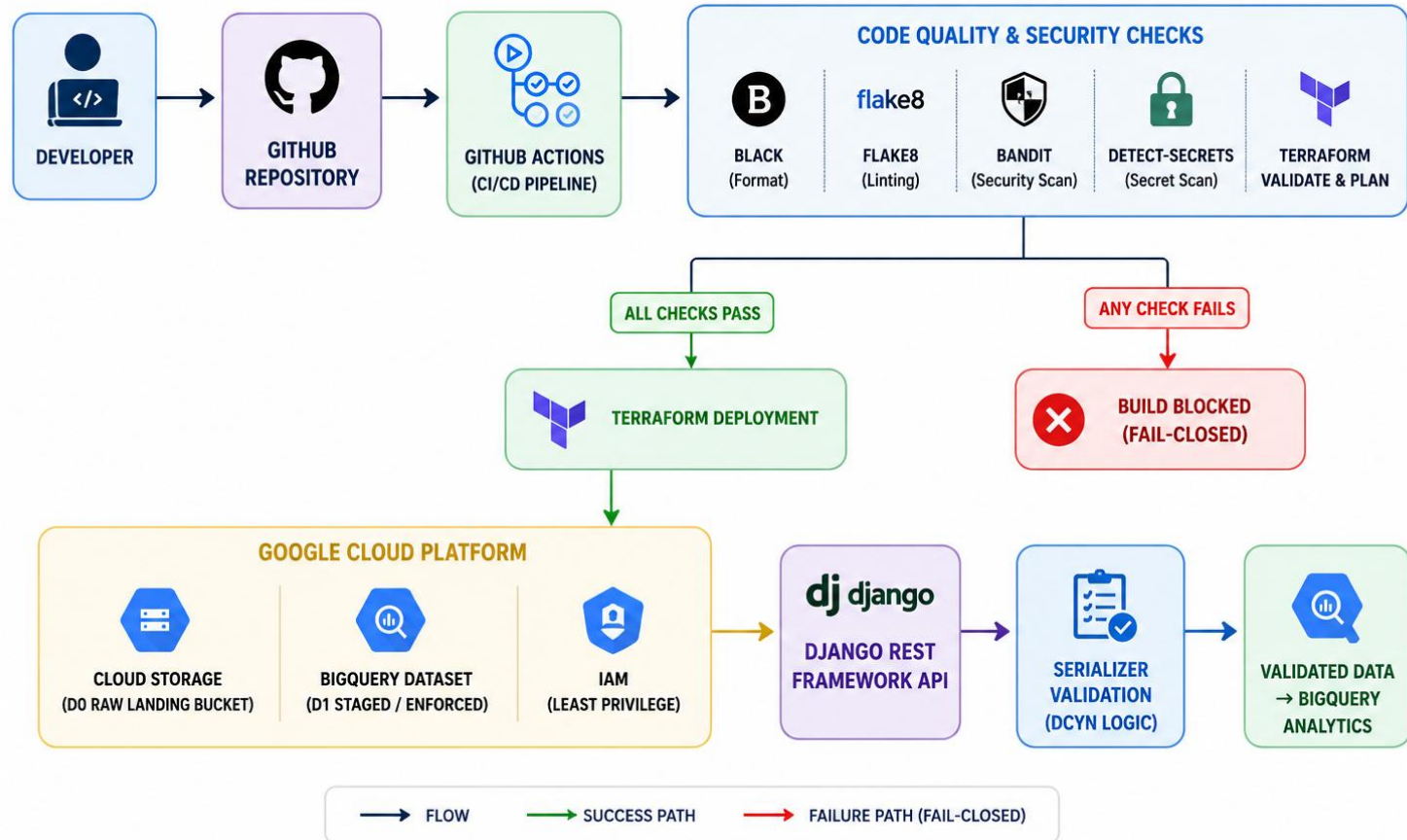
- Infrastructure as Code
 - Secure IAM & Least Privilege
 - Row-Level Security
 - Automated validation
- 
- 

TECHNOLOGY STACK

- Terraform
 - Google Cloud Platform
 - GitHub Actions
 - Django REST Framework
 - BigQuery
 - Cloud Storage

PROJECT ARCHITECTURE

PROJECT ARCHITECTURE





TASK 1: TERRAFORM

- • Provider & Variables
 - GCS Raw Landing Bucket
 - BigQuery Dataset
 - IAM
 - Public Access Prevention
 - Versioning
- 
- 
- 

SECURITY IMPLEMENTATION

- • IAM (RBAC)
 - Principle of Least Privilege
 - Row-Level Security (RLS)
 - Secret Protection

TASK 2: CI/CD

- Pipeline:

Checkout → terraform fmt → validate → flake8 →
Secret Scan → Deploy

Successful pipeline

The screenshot displays the GitHub Actions interface for a workflow named 'Habot Secure CI Pipeline'. The workflow is triggered by a push to the 'main' branch. The job 'security-checks' has completed successfully, with a duration of 14 seconds. The interface includes a sidebar with navigation options like 'Summary', 'All jobs', 'security-checks', 'Run details', 'Usage', and 'Workflow file'. The main area shows the job status as 'Success' and provides details about the trigger and duration.

Summary

security-checks

Run details

Usage

Workflow file

← Habot Secure CI Pipeline

✓ updated main.tf #39

Re-run all jobs

Triggered via push now

SuryavanshiAmruta pushed -> 606b0f7 main

Status: Success

Total duration: 23s

Artifacts: -

ci.yml
on: push

✓ security-checks 14s

FAIL-CLOSED BUILD GATE

- Pipeline immediately stops if:
 - Formatting fails
 - Linting fails
 - Hardcoded API key detected
 - Terraform validation fails

TASK 3: DRF VALIDATION

- • Student onboarding JSON
- Serializer validation
- DCYN Yes/No logic
- Strict field validation

TESTING & RESULTS

- Terraform Apply Successful
- Bucket & Dataset Created
- IAM Verified
- Pipeline Tested
- API Validation Successful

Terraform Apply

```

google_bigquery_dataset.staging: Creation complete after 3s [id=projects/habot-devops-project-504219/datasets/d1_staged]
google_storage_bucket.raw_landing: Creation complete after 4s [id=habot-raw-landing-504219]
google_storage_bucket_iam_member.viewer: Creating...
google_storage_bucket_iam_member.viewer: Creation complete after 6s [id=b/habot-raw-landing-504219/roles/storage.objectViewer/user:amrutasuryavanshi212@gmail.com]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

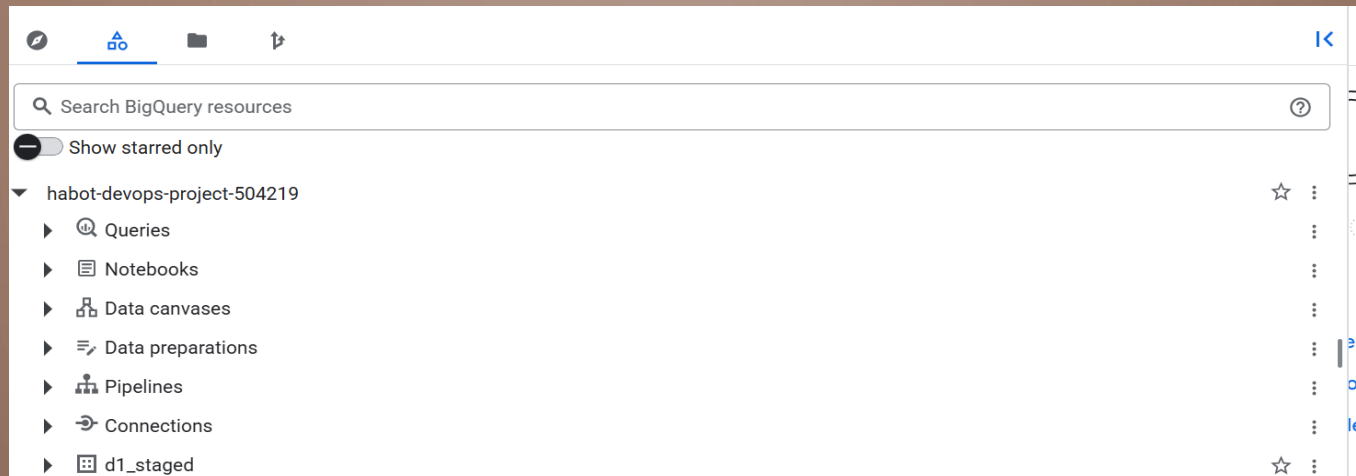
bucket_name = "habot-raw-landing-504219"
dataset_name = "d1_staged"
PS D:\Devops_projects\habot-devops-project\terraform>

```

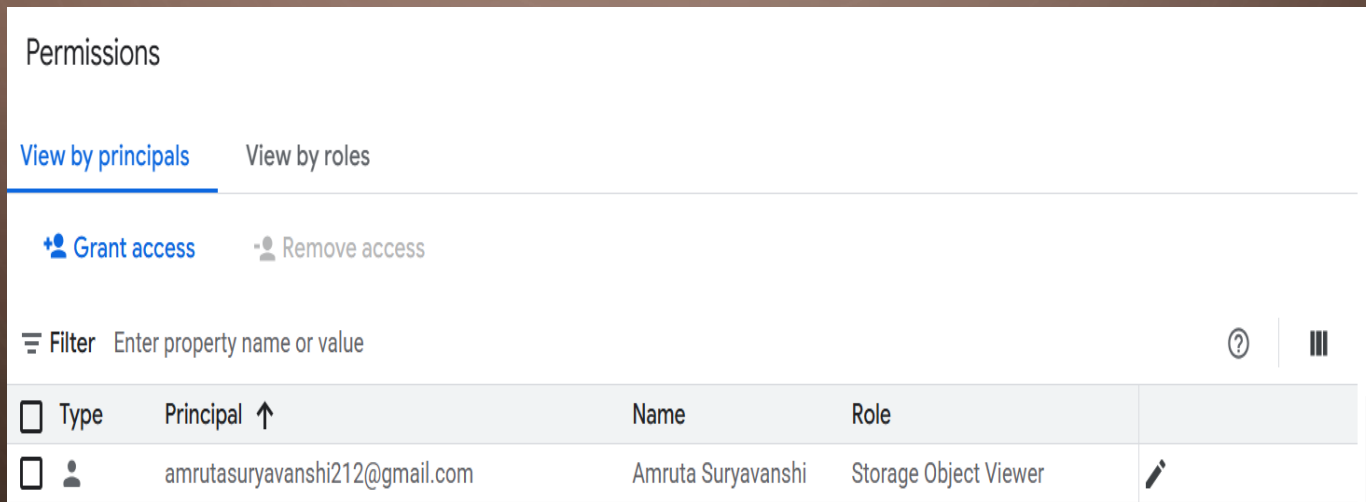
Storage bucket

Buckets						+ Create	Refresh	Go to path	Learn
Filter Filter buckets									
☐	Name ↑	Created	Location type	Location	Default storage class ?	Last modified			
☐	habot-raw-landing-504219	Aug 2, 2026, 1:41:59 AM	Multi-region	asia	Standard	Aug 2, 2026, 1:41:59 AM			

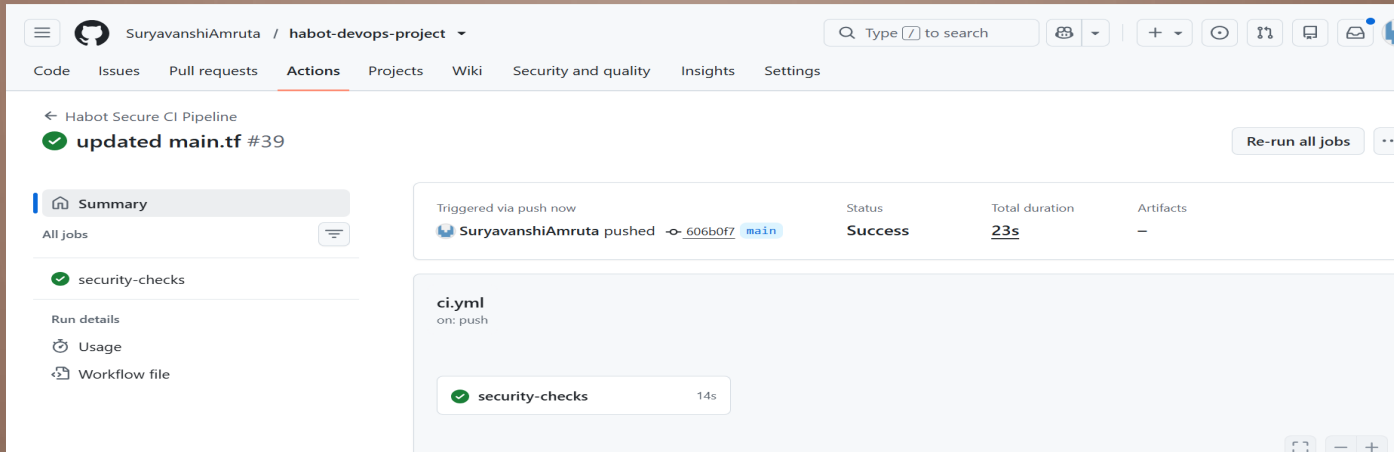
BigQuery dataset



IAM Verified



Pipeline Tested



The screenshot shows a GitHub Actions workflow run for the repository 'SuryavanshiAmruta / habot-devops-project'. The workflow is named 'Habot Secure CI Pipeline' and the specific run is 'updated main.tf #39'. The run status is 'Success'. The workflow file is 'ci.yml' and it was triggered by a push to the 'main' branch. The total duration of the run is 23 seconds. The workflow consists of a single job named 'security-checks' which took 14 seconds to complete. The job details show that it was triggered via a push now and the status is 'Success'.

Summary

All jobs

- security-checks

Run details

- Usage
- Workflow file

Triggered via push now

SuryavanshiAmruta pushed · 606b0f7 · main

Status: Success

Total duration: 23s

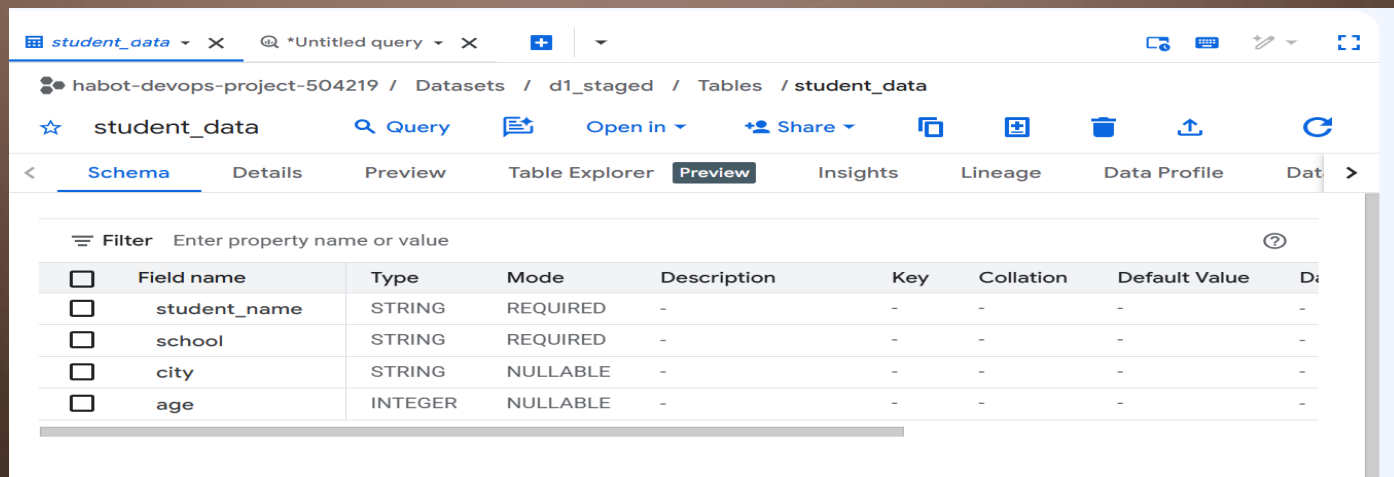
Artifacts: -

ci.yml

on: push

security-checks 14s

API Validation Successful



The screenshot shows the Databricks Table Explorer interface for the table 'student_data' in the 'habot-devops-project-504219' workspace. The table is located in the 'd1_staged' dataset. The schema is displayed in a table format with columns: Field name, Type, Mode, Description, Key, Collation, Default Value, and Data. The fields are 'student_name', 'school', 'city', and 'age'. The 'student_name' and 'school' fields are required (REQUIRED), while 'city' and 'age' are nullable (NULLABLE). The 'city' field is a string (STRING) and the 'age' field is an integer (INTEGER).

student_oata x *Untitled query x

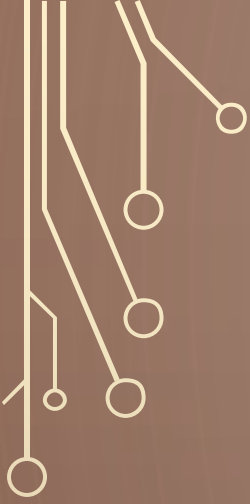
habot-devops-project-504219 / Datasets / d1_staged / Tables / student_data

student_data Query Open in Share

Schema Details Preview Table Explorer Preview Insights Lineage Data Profile Data

Filter Enter property name or value

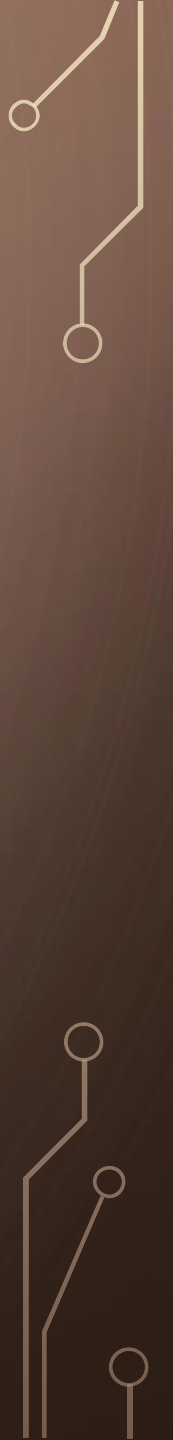
Field name	Type	Mode	Description	Key	Collation	Default Value	Data
student_name	STRING	REQUIRED	-	-	-	-	-
school	STRING	REQUIRED	-	-	-	-	-
city	STRING	NULLABLE	-	-	-	-	-
age	INTEGER	NULLABLE	-	-	-	-	-



CHALLENGES

- Learning IAM & RLS
- CI/CD integration
- Cloud security

FUTURE IMPROVEMENTS

- Remote Terraform state
 - Secret Manager
 - Cloud Monitoring
 - Multi-environment deployment
- 
- 

CONCLUSION

- Project meets HaborConnect requirements using secure IaC, automated CI/CD, robust validation, and cloud security best practices.

The background is a dark brown gradient with faint, concentric circles centered around the text. In the corners, there are stylized, light-colored lines resembling circuit traces or a network diagram, with small circles at the end of the lines.

THANK YOU